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Abstract:

The ever-increasing numbers of new wireless technologies and services have resulted in increasing demand of smart wireless devices. The smart antennas in such devices should support all the required specifications so that these are compatible with different applications. Moreover, for realizing emerging wireless application such as cognitive radios, new requirements on the antenna are imposed. Novel antenna solutions which can easily reconfigure its operating parameter like frequency, bandwidth, pattern or polarization are required to support multiband, multimode and multi standard applications. Such antennas must support all these features in a single unit to produce small size, low cost, low profile integrated antenna which can be easily incorporated into the feature-rich applications. This paper deals with the design, simulation and fabrication of frequency reconfigurable antenna which have more than one reconfiguration features. In these we are adding extended parasitic elements to the Co-planar mono pole antenna for more than one resonant frequency. Co-planar waveguide fed polarization reconfigurable monopole antenna is used and PIN diodes are used as the switching device. The relevant design techniques, simulation results and fabrication details are presented. The integrated antenna structures need to be experimentally tested to validate the established parameters for antenna performance such as return loss, radiation pattern and gain.

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